

Full evaluation results

Every run x every evaluation seed, matrix-v2 batch (ppo / ctrl / cppo) + the cppo15 arm. One table per run; columns are the three evaluation seeds plus their arithmetic mean.

Provenance.

Task: Isaac-Lift-Cube-UR5e-v0 (frozen weld env).

Training: ppo/ctrl/cppo -- commit 567e4c0, tag matrix-v2, 1500 iterations, num_envs=4096, seeds 1-5 and 50-54. cppo15 -- commit 866ea33, tag matrix-v2-cppo15, same protocol, cost_limit=15 (differs from cppo by that one field only).

Evaluation: eval_policy.py, 1000 episodes per (checkpoint x eval-seed), num_envs=128, eval seeds 101/102/103 (disjoint from training seeds), deterministic policy, lift_frac=0.50, goal tolerances 1/2/5 cm.

Source: ur5_grasp/tools/eval_policy_results.csv, resolved to the newest-dated checkpoint per (label, eval_seed) -- this file is append-only and carries stale rows from earlier sweeps under the same labels; 18 of the 120 combinations used in this report had a stale duplicate that was excluded by checkpoint date.

Generated by results/scripts/make_full_eval_pdf.py.

Reading the tables. Two metrics are marked <- headline: lift success (object reaches at least half the commanded goal height, at the end of the episode) and goal-reach @ 1 cm, per the audit's reporting rule (a single threshold like the legacy 5 cm bar saturates near 0% or 100% once a policy converges, so it cannot discriminate two converged policies -- report the full distribution instead). "Step" safety metrics are soft-margin fractions over every control step across all 1000 episodes; "episode" safety metrics are the percentage of episodes that touched that constraint at all, i.e. $\text{sing_frac}/\text{joint_frac}/\text{coll_frac} > 0$ for at least one step.

ppo (unconstrained PPO baseline)

ppo_s1 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success $\geq 0.50 \times$ commanded goal height (%) <- headline	100.00	99.70	99.80	99.83
... bar reached at any point in the episode (%)	100.00	99.70	99.80	99.83
... legacy fixed-height definition, Day-22 (%)	100.00	99.70	99.80	99.83
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	80.10	80.20	79.70	80.00
Goal-reach success @ 2 cm (%)	97.40	97.30	96.30	97.00
Goal-reach success @ 5 cm (%)	100.00	99.70	99.70	99.80
Final goal distance -- mean (m)	0.0079	0.0093	0.0087	0.0086
Final goal distance -- median (m)	0.0065	0.0066	0.0066	0.0066
Final goal distance -- p90 (m)	0.0130	0.0131	0.0130	0.0130
Final goal distance -- worst episode (m)	0.0417	0.5969	0.4236	0.3541
Singularity margin violated (% of steps)	22.88	25.16	23.57	23.87
Joint-limit margin violated (% of steps)	37.10	34.27	35.73	35.70
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	48.60	49.10	48.10	48.60
Episodes that touched the joint-limit margin (%)	49.60	46.50	47.40	47.83
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0574	0.0568	0.0573	0.0572
Manipulability w -- worst single episode	0.0036	0.0077	0.0034	0.0049
Episodic safety cost -- mean	107.41	101.74	104.42	104.52
Episodic safety cost -- p90	203.73	202.36	203.97	203.35

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift/2026-08-01_00-01-25_ppo_s1/model_1499.pt

ppo_s2 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.60	99.60	99.60	99.60
... bar reached at any point in the episode (%)	99.60	99.60	99.60	99.60
... legacy fixed-height definition, Day-22 (%)	99.60	99.60	99.60	99.60
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	97.40	97.40	97.10	97.30
Goal-reach success @ 2 cm (%)	99.60	99.60	99.60	99.60
Goal-reach success @ 5 cm (%)	99.60	99.60	99.60	99.60
Final goal distance -- mean (m)	0.0067	0.0068	0.0063	0.0066
Final goal distance -- median (m)	0.0046	0.0045	0.0045	0.0045
Final goal distance -- p90 (m)	0.0071	0.0072	0.0070	0.0071
Final goal distance -- worst episode (m)	0.5631	0.6439	0.4313	0.5461
Singularity margin violated (% of steps)	12.59	13.65	13.72	13.32
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	37.60	39.30	38.40	38.43
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.10	0.03
Manipulability w -- mean of episode-minimum	0.0628	0.0622	0.0630	0.0627
Manipulability w -- worst single episode	0.0000	0.0001	0.0001	0.0001
Episodic safety cost -- mean	7.99	7.71	8.45	8.05
Episodic safety cost -- p90	24.29	27.56	28.54	26.80

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift/2026-08-01_00-13-17_ppo_s2/model_1499.pt

ppo_s3 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	249.9	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	100.00	99.90	100.00	99.97
... bar reached at any point in the episode (%)	100.00	99.90	100.00	99.97
... legacy fixed-height definition, Day-22 (%)	100.00	99.90	100.00	99.97
Mean commanded goal height (m)	0.296	0.295	0.300	0.297
Goal-reach success @ 1 cm (%) <- headline	98.70	99.00	98.60	98.77
Goal-reach success @ 2 cm (%)	99.60	99.50	99.50	99.53
Goal-reach success @ 5 cm (%)	100.00	99.80	100.00	99.93
Final goal distance -- mean (m)	0.0052	0.0061	0.0053	0.0055
Final goal distance -- median (m)	0.0050	0.0051	0.0051	0.0051
Final goal distance -- p90 (m)	0.0071	0.0070	0.0070	0.0070
Final goal distance -- worst episode (m)	0.0241	0.8120	0.0279	0.2880
Singularity margin violated (% of steps)	91.07	91.43	91.60	91.37
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	99.00	99.30	99.50	99.27
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.10	0.00	0.03
Manipulability w -- mean of episode-minimum	0.0118	0.0109	0.0102	0.0110
Manipulability w -- worst single episode	0.0000	0.0000	0.0000	0.0000
Episodic safety cost -- mean	161.98	165.23	166.43	164.55
Episodic safety cost -- p90	210.81	213.05	214.17	212.68

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift/2026-08-01_00-24-40_ppo_s3/model_1499.pt

ppo_s4 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	100.00	100.00	100.00	100.00
... bar reached at any point in the episode (%)	100.00	100.00	100.00	100.00
... legacy fixed-height definition, Day-22 (%)	100.00	100.00	100.00	100.00
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	94.20	95.20	95.10	94.83
Goal-reach success @ 2 cm (%)	98.70	99.10	98.60	98.80
Goal-reach success @ 5 cm (%)	100.00	100.00	100.00	100.00
Final goal distance -- mean (m)	0.0050	0.0048	0.0050	0.0049
Final goal distance -- median (m)	0.0043	0.0042	0.0043	0.0043
Final goal distance -- p90 (m)	0.0072	0.0074	0.0073	0.0073
Final goal distance -- worst episode (m)	0.0409	0.0335	0.0366	0.0370
Singularity margin violated (% of steps)	37.49	39.58	38.17	38.41
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	43.70	45.50	43.70	44.30
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0613	0.0598	0.0612	0.0608
Manipulability w -- worst single episode	0.0000	0.0023	0.0000	0.0008
Episodic safety cost -- mean	30.36	33.77	30.62	31.58
Episodic safety cost -- p90	109.57	118.41	110.63	112.87

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift/2026-08-01_00-35-41_ppo_s4/model_1499.pt

ppo_s5 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	249.9	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	99.80	99.60	99.77
... bar reached at any point in the episode (%)	99.90	99.80	99.60	99.77
... legacy fixed-height definition, Day-22 (%)	99.90	99.80	99.60	99.77
Mean commanded goal height (m)	0.296	0.296	0.300	0.297
Goal-reach success @ 1 cm (%) <- headline	93.10	93.40	94.50	93.67
Goal-reach success @ 2 cm (%)	99.40	99.10	99.20	99.23
Goal-reach success @ 5 cm (%)	99.90	99.80	99.60	99.77
Final goal distance -- mean (m)	0.0060	0.0068	0.0066	0.0065
Final goal distance -- median (m)	0.0045	0.0045	0.0045	0.0045
Final goal distance -- p90 (m)	0.0085	0.0087	0.0081	0.0084
Final goal distance -- worst episode (m)	0.5884	1.0516	0.4168	0.6856
Singularity margin violated (% of steps)	24.69	26.64	26.19	25.84
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	35.30	37.10	36.30	36.23
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0632	0.0628	0.0623	0.0628
Manipulability w -- worst single episode	0.0071	0.0073	0.0080	0.0075
Episodic safety cost -- mean	18.71	18.57	19.97	19.08
Episodic safety cost -- p90	84.35	80.40	83.23	82.66

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift/2026-08-01_00-46-59_ppo_s5/model_1499.pt

ppo_s50 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	249.8	250.0	250.0	249.9
Lift success >= 0.50 x commanded goal height (%) <- headline	99.80	99.90	100.00	99.90
... bar reached at any point in the episode (%)	99.80	99.90	100.00	99.90
... legacy fixed-height definition, Day-22 (%)	99.80	99.90	100.00	99.90
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	89.30	87.50	88.70	88.50
Goal-reach success @ 2 cm (%)	98.10	98.20	98.60	98.30
Goal-reach success @ 5 cm (%)	99.80	99.90	100.00	99.90
Final goal distance -- mean (m)	0.0071	0.0065	0.0058	0.0065
Final goal distance -- median (m)	0.0045	0.0046	0.0046	0.0046
Final goal distance -- p90 (m)	0.0104	0.0115	0.0109	0.0109
Final goal distance -- worst episode (m)	0.9080	0.6413	0.0472	0.5322
Singularity margin violated (% of steps)	14.74	14.85	13.16	14.25
Joint-limit margin violated (% of steps)	0.10	0.00	0.01	0.04
Collision floor violated (% of steps)	0.03	0.02	0.02	0.02
Episodes that touched the singularity margin (%)	50.20	50.80	46.70	49.23
Episodes that touched the joint-limit margin (%)	0.30	0.10	0.10	0.17
Episodes that touched the collision floor (%)	1.20	0.70	1.50	1.13
Manipulability w -- mean of episode-minimum	0.0581	0.0571	0.0585	0.0579
Manipulability w -- worst single episode	0.0001	0.0004	0.0003	0.0003
Episodic safety cost -- mean	9.36	10.17	7.23	8.92
Episodic safety cost -- p90	20.91	21.86	15.26	19.34

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift/2026-08-01_04-12-13_ppo_s50/model_1499.pt

ppo_s51 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	100.00	100.00	99.97
... bar reached at any point in the episode (%)	99.90	100.00	100.00	99.97
... legacy fixed-height definition, Day-22 (%)	99.90	100.00	100.00	99.97
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.10	99.50	99.40	99.33
Goal-reach success @ 2 cm (%)	99.90	100.00	100.00	99.97
Goal-reach success @ 5 cm (%)	99.90	100.00	100.00	99.97
Final goal distance -- mean (m)	0.0048	0.0043	0.0042	0.0044
Final goal distance -- median (m)	0.0041	0.0041	0.0041	0.0041
Final goal distance -- p90 (m)	0.0060	0.0058	0.0057	0.0058
Final goal distance -- worst episode (m)	0.5130	0.0132	0.0114	0.1792
Singularity margin violated (% of steps)	7.21	8.58	7.60	7.80
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	22.00	24.20	21.10	22.43
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0707	0.0703	0.0712	0.0707
Manipulability w -- worst single episode	0.0320	0.0368	0.0316	0.0335
Episodic safety cost -- mean	1.58	2.06	1.82	1.82
Episodic safety cost -- p90	2.24	4.26	3.31	3.27

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift/2026-08-01_04-23-37_ppo_s51/model_1499.pt

ppo_s52 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	249.8	249.9
Lift success >= 0.50 x commanded goal height (%) <- headline	100.00	100.00	99.80	99.93
... bar reached at any point in the episode (%)	100.00	100.00	99.90	99.97
... legacy fixed-height definition, Day-22 (%)	100.00	100.00	99.80	99.93
Mean commanded goal height (m)	0.296	0.299	0.299	0.298
Goal-reach success @ 1 cm (%) <- headline	93.90	95.30	93.30	94.17
Goal-reach success @ 2 cm (%)	98.90	98.90	98.50	98.77
Goal-reach success @ 5 cm (%)	99.50	99.50	99.60	99.53
Final goal distance -- mean (m)	0.0052	0.0050	0.0062	0.0055
Final goal distance -- median (m)	0.0041	0.0040	0.0041	0.0041
Final goal distance -- p90 (m)	0.0076	0.0075	0.0085	0.0079
Final goal distance -- worst episode (m)	0.0715	0.0840	0.7909	0.3155
Singularity margin violated (% of steps)	67.12	69.81	66.95	67.96
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	75.70	78.90	75.80	76.80
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0317	0.0294	0.0309	0.0307
Manipulability w -- worst single episode	0.0000	0.0000	0.0000	0.0000
Episodic safety cost -- mean	106.24	112.27	109.71	109.41
Episodic safety cost -- p90	204.26	207.21	206.09	205.85

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift/2026-08-01_04-34-25_ppo_s52/model_1499.pt

ppo_s53 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	100.00	99.50	99.80
... bar reached at any point in the episode (%)	99.90	100.00	99.60	99.83
... legacy fixed-height definition, Day-22 (%)	99.90	100.00	99.50	99.80
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.10	98.90	98.40	98.80
Goal-reach success @ 2 cm (%)	99.90	100.00	99.40	99.77
Goal-reach success @ 5 cm (%)	99.90	100.00	99.50	99.80
Final goal distance -- mean (m)	0.0050	0.0045	0.0068	0.0054
Final goal distance -- median (m)	0.0044	0.0044	0.0043	0.0044
Final goal distance -- p90 (m)	0.0067	0.0067	0.0068	0.0067
Final goal distance -- worst episode (m)	0.4654	0.0196	0.6451	0.3767
Singularity margin violated (% of steps)	30.06	32.86	30.38	31.10
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	44.10	46.10	44.60	44.93
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0611	0.0600	0.0610	0.0607
Manipulability w -- worst single episode	0.0004	0.0057	0.0001	0.0021
Episodic safety cost -- mean	17.88	19.71	18.29	18.63
Episodic safety cost -- p90	74.36	72.13	72.94	73.14

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift/2026-08-01_04-45-32_ppo_s53/model_1499.pt

ppo_s54 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	99.70	99.80	99.80
... bar reached at any point in the episode (%)	99.90	99.90	99.90	99.90
... legacy fixed-height definition, Day-22 (%)	99.90	99.70	99.80	99.80
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	97.50	97.30	97.40	97.40
Goal-reach success @ 2 cm (%)	99.90	99.70	99.80	99.80
Goal-reach success @ 5 cm (%)	99.90	99.70	99.80	99.80
Final goal distance -- mean (m)	0.0057	0.0067	0.0061	0.0062
Final goal distance -- median (m)	0.0051	0.0052	0.0052	0.0052
Final goal distance -- p90 (m)	0.0073	0.0072	0.0071	0.0072
Final goal distance -- worst episode (m)	0.3725	0.5497	0.4180	0.4467
Singularity margin violated (% of steps)	8.05	7.36	7.66	7.69
Joint-limit margin violated (% of steps)	4.15	4.75	3.84	4.25
Collision floor violated (% of steps)	0.00	0.05	0.00	0.02
Episodes that touched the singularity margin (%)	17.20	16.60	17.60	17.13
Episodes that touched the joint-limit margin (%)	5.50	6.30	5.20	5.67
Episodes that touched the collision floor (%)	0.00	0.30	0.10	0.13
Manipulability w -- mean of episode-minimum	0.0729	0.0728	0.0726	0.0728
Manipulability w -- worst single episode	0.0119	0.0230	0.0111	0.0153
Episodic safety cost -- mean	10.39	10.35	9.86	10.20
Episodic safety cost -- p90	3.07	3.11	2.66	2.95

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift/2026-08-01_04-56-17_ppo_s54/model_1499.pt

ctrl (cost critic present, lambda pinned to 0 -- the control arm)

ctrl_s1 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success $\geq 0.50 \times$ commanded goal height (%) <- headline	100.00	99.70	99.80	99.83
... bar reached at any point in the episode (%)	100.00	99.70	99.80	99.83
... legacy fixed-height definition, Day-22 (%)	100.00	99.70	99.80	99.83
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	80.10	80.20	79.70	80.00
Goal-reach success @ 2 cm (%)	97.40	97.30	96.30	97.00
Goal-reach success @ 5 cm (%)	100.00	99.70	99.70	99.80
Final goal distance -- mean (m)	0.0079	0.0093	0.0087	0.0086
Final goal distance -- median (m)	0.0065	0.0066	0.0066	0.0066
Final goal distance -- p90 (m)	0.0130	0.0131	0.0130	0.0130
Final goal distance -- worst episode (m)	0.0417	0.5969	0.4236	0.3541
Singularity margin violated (% of steps)	22.88	25.16	23.57	23.87
Joint-limit margin violated (% of steps)	37.10	34.27	35.73	35.70
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	48.60	49.10	48.10	48.60
Episodes that touched the joint-limit margin (%)	49.60	46.50	47.40	47.83
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0574	0.0568	0.0573	0.0572
Manipulability w -- worst single episode	0.0036	0.0077	0.0034	0.0049
Episodic safety cost -- mean	107.41	101.74	104.42	104.52
Episodic safety cost -- p90	203.73	202.36	203.97	203.35

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs1_rl/ur5e_lift_ctrl/2026-08-01_00-58-26_ctrl_s1/model_1499.pt

ctrl_s2 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.60	99.60	99.60	99.60
... bar reached at any point in the episode (%)	99.60	99.60	99.60	99.60
... legacy fixed-height definition, Day-22 (%)	99.60	99.60	99.60	99.60
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	97.40	97.40	97.10	97.30
Goal-reach success @ 2 cm (%)	99.60	99.60	99.60	99.60
Goal-reach success @ 5 cm (%)	99.60	99.60	99.60	99.60
Final goal distance -- mean (m)	0.0067	0.0068	0.0063	0.0066
Final goal distance -- median (m)	0.0046	0.0045	0.0045	0.0045
Final goal distance -- p90 (m)	0.0071	0.0072	0.0070	0.0071
Final goal distance -- worst episode (m)	0.5631	0.6439	0.4313	0.5461
Singularity margin violated (% of steps)	12.59	13.65	13.72	13.32
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	37.60	39.30	38.40	38.43
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.10	0.03
Manipulability w -- mean of episode-minimum	0.0628	0.0622	0.0630	0.0627
Manipulability w -- worst single episode	0.0000	0.0001	0.0001	0.0001
Episodic safety cost -- mean	7.99	7.71	8.45	8.05
Episodic safety cost -- p90	24.29	27.56	28.54	26.80

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_01-10-06_ctrl_s2/model_1499.pt

ctrl_s3 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	249.9	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	100.00	99.90	100.00	99.97
... bar reached at any point in the episode (%)	100.00	99.90	100.00	99.97
... legacy fixed-height definition, Day-22 (%)	100.00	99.90	100.00	99.97
Mean commanded goal height (m)	0.296	0.295	0.300	0.297
Goal-reach success @ 1 cm (%) <- headline	98.70	99.00	98.60	98.77
Goal-reach success @ 2 cm (%)	99.60	99.50	99.50	99.53
Goal-reach success @ 5 cm (%)	100.00	99.80	100.00	99.93
Final goal distance -- mean (m)	0.0052	0.0061	0.0053	0.0055
Final goal distance -- median (m)	0.0050	0.0051	0.0051	0.0051
Final goal distance -- p90 (m)	0.0071	0.0070	0.0070	0.0070
Final goal distance -- worst episode (m)	0.0241	0.8120	0.0279	0.2880
Singularity margin violated (% of steps)	91.07	91.43	91.60	91.37
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	99.00	99.30	99.50	99.27
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.10	0.00	0.03
Manipulability w -- mean of episode-minimum	0.0118	0.0109	0.0102	0.0110
Manipulability w -- worst single episode	0.0000	0.0000	0.0000	0.0000
Episodic safety cost -- mean	161.98	165.23	166.43	164.55
Episodic safety cost -- p90	210.81	213.05	214.17	212.68

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_01-21-49_ctrl_s3/model_1499.pt

ctrl_s4 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success $\geq 0.50 \times$ commanded goal height (%) <- headline	100.00	100.00	100.00	100.00
... bar reached at any point in the episode (%)	100.00	100.00	100.00	100.00
... legacy fixed-height definition, Day-22 (%)	100.00	100.00	100.00	100.00
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	94.20	95.20	95.10	94.83
Goal-reach success @ 2 cm (%)	98.70	99.10	98.60	98.80
Goal-reach success @ 5 cm (%)	100.00	100.00	100.00	100.00
Final goal distance -- mean (m)	0.0050	0.0048	0.0050	0.0049
Final goal distance -- median (m)	0.0043	0.0042	0.0043	0.0043
Final goal distance -- p90 (m)	0.0072	0.0074	0.0073	0.0073
Final goal distance -- worst episode (m)	0.0409	0.0335	0.0366	0.0370
Singularity margin violated (% of steps)	37.49	39.58	38.17	38.41
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	43.70	45.50	43.70	44.30
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0613	0.0598	0.0612	0.0608
Manipulability w -- worst single episode	0.0000	0.0023	0.0000	0.0008
Episodic safety cost -- mean	30.36	33.77	30.62	31.58
Episodic safety cost -- p90	109.57	118.41	110.63	112.87

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_01-33-16_ctrl_s4/model_1499.pt

ctrl_s5 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	249.9	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	99.80	99.60	99.77
... bar reached at any point in the episode (%)	99.90	99.80	99.60	99.77
... legacy fixed-height definition, Day-22 (%)	99.90	99.80	99.60	99.77
Mean commanded goal height (m)	0.296	0.296	0.300	0.297
Goal-reach success @ 1 cm (%) <- headline	93.10	93.40	94.50	93.67
Goal-reach success @ 2 cm (%)	99.40	99.10	99.20	99.23
Goal-reach success @ 5 cm (%)	99.90	99.80	99.60	99.77
Final goal distance -- mean (m)	0.0060	0.0068	0.0066	0.0065
Final goal distance -- median (m)	0.0045	0.0045	0.0045	0.0045
Final goal distance -- p90 (m)	0.0085	0.0087	0.0081	0.0084
Final goal distance -- worst episode (m)	0.5884	1.0516	0.4168	0.6856
Singularity margin violated (% of steps)	24.69	26.64	26.19	25.84
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	35.30	37.10	36.30	36.23
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0632	0.0628	0.0623	0.0628
Manipulability w -- worst single episode	0.0071	0.0073	0.0080	0.0075
Episodic safety cost -- mean	18.71	18.57	19.97	19.08
Episodic safety cost -- p90	84.35	80.40	83.23	82.66

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_01-45-10_ctrl_s5/model_1499.pt

ctrl_s50 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	249.8	250.0	250.0	249.9
Lift success >= 0.50 x commanded goal height (%) <- headline	99.80	99.90	100.00	99.90
... bar reached at any point in the episode (%)	99.80	99.90	100.00	99.90
... legacy fixed-height definition, Day-22 (%)	99.80	99.90	100.00	99.90
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	89.30	87.50	88.70	88.50
Goal-reach success @ 2 cm (%)	98.10	98.20	98.60	98.30
Goal-reach success @ 5 cm (%)	99.80	99.90	100.00	99.90
Final goal distance -- mean (m)	0.0071	0.0065	0.0058	0.0065
Final goal distance -- median (m)	0.0045	0.0046	0.0046	0.0046
Final goal distance -- p90 (m)	0.0104	0.0115	0.0109	0.0109
Final goal distance -- worst episode (m)	0.9080	0.6413	0.0472	0.5322
Singularity margin violated (% of steps)	14.74	14.85	13.16	14.25
Joint-limit margin violated (% of steps)	0.10	0.00	0.01	0.04
Collision floor violated (% of steps)	0.03	0.02	0.02	0.02
Episodes that touched the singularity margin (%)	50.20	50.80	46.70	49.23
Episodes that touched the joint-limit margin (%)	0.30	0.10	0.10	0.17
Episodes that touched the collision floor (%)	1.20	0.70	1.50	1.13
Manipulability w -- mean of episode-minimum	0.0581	0.0571	0.0585	0.0579
Manipulability w -- worst single episode	0.0001	0.0004	0.0003	0.0003
Episodic safety cost -- mean	9.36	10.17	7.23	8.92
Episodic safety cost -- p90	20.91	21.86	15.26	19.34

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_05-07-30_ctrl_s50/model_1499.pt

ctrl_s51 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	100.00	100.00	99.97
... bar reached at any point in the episode (%)	99.90	100.00	100.00	99.97
... legacy fixed-height definition, Day-22 (%)	99.90	100.00	100.00	99.97
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.10	99.50	99.40	99.33
Goal-reach success @ 2 cm (%)	99.90	100.00	100.00	99.97
Goal-reach success @ 5 cm (%)	99.90	100.00	100.00	99.97
Final goal distance -- mean (m)	0.0048	0.0043	0.0042	0.0044
Final goal distance -- median (m)	0.0041	0.0041	0.0041	0.0041
Final goal distance -- p90 (m)	0.0060	0.0058	0.0057	0.0058
Final goal distance -- worst episode (m)	0.5130	0.0132	0.0114	0.1792
Singularity margin violated (% of steps)	7.21	8.58	7.60	7.80
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	22.00	24.20	21.10	22.43
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0707	0.0703	0.0712	0.0707
Manipulability w -- worst single episode	0.0320	0.0368	0.0316	0.0335
Episodic safety cost -- mean	1.58	2.06	1.82	1.82
Episodic safety cost -- p90	2.24	4.26	3.31	3.27

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_05-19-17_ctrl_s51/model_1499.pt

ctrl_s52 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	249.8	249.9
Lift success $\geq 0.50 \times$ commanded goal height (%) <- headline	100.00	100.00	99.80	99.93
... bar reached at any point in the episode (%)	100.00	100.00	99.90	99.97
... legacy fixed-height definition, Day-22 (%)	100.00	100.00	99.80	99.93
Mean commanded goal height (m)	0.296	0.299	0.299	0.298
Goal-reach success @ 1 cm (%) <- headline	93.90	95.30	93.30	94.17
Goal-reach success @ 2 cm (%)	98.90	98.90	98.50	98.77
Goal-reach success @ 5 cm (%)	99.50	99.50	99.60	99.53
Final goal distance -- mean (m)	0.0052	0.0050	0.0062	0.0055
Final goal distance -- median (m)	0.0041	0.0040	0.0041	0.0041
Final goal distance -- p90 (m)	0.0076	0.0075	0.0085	0.0079
Final goal distance -- worst episode (m)	0.0715	0.0840	0.7909	0.3155
Singularity margin violated (% of steps)	67.12	69.81	66.95	67.96
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	75.70	78.90	75.80	76.80
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0317	0.0294	0.0309	0.0307
Manipulability w -- worst single episode	0.0000	0.0000	0.0000	0.0000
Episodic safety cost -- mean	106.24	112.27	109.71	109.41
Episodic safety cost -- p90	204.26	207.21	206.09	205.85

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_05-30-34_ctrl_s52/model_1499.pt

ctrl_s53 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success ≥ 0.50 x commanded goal height (%) <- headline	99.90	100.00	99.50	99.80
... bar reached at any point in the episode (%)	99.90	100.00	99.60	99.83
... legacy fixed-height definition, Day-22 (%)	99.90	100.00	99.50	99.80
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.10	98.90	98.40	98.80
Goal-reach success @ 2 cm (%)	99.90	100.00	99.40	99.77
Goal-reach success @ 5 cm (%)	99.90	100.00	99.50	99.80
Final goal distance -- mean (m)	0.0050	0.0045	0.0068	0.0054
Final goal distance -- median (m)	0.0044	0.0044	0.0043	0.0044
Final goal distance -- p90 (m)	0.0067	0.0067	0.0068	0.0067
Final goal distance -- worst episode (m)	0.4654	0.0196	0.6451	0.3767
Singularity margin violated (% of steps)	30.06	32.86	30.38	31.10
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	44.10	46.10	44.60	44.93
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0611	0.0600	0.0610	0.0607
Manipulability w -- worst single episode	0.0004	0.0057	0.0001	0.0021
Episodic safety cost -- mean	17.88	19.71	18.29	18.63
Episodic safety cost -- p90	74.36	72.13	72.94	73.14

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_05-42-05_ctrl_s53/model_1499.pt

ctrl_s54 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	99.70	99.80	99.80
... bar reached at any point in the episode (%)	99.90	99.90	99.90	99.90
... legacy fixed-height definition, Day-22 (%)	99.90	99.70	99.80	99.80
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	97.50	97.30	97.40	97.40
Goal-reach success @ 2 cm (%)	99.90	99.70	99.80	99.80
Goal-reach success @ 5 cm (%)	99.90	99.70	99.80	99.80
Final goal distance -- mean (m)	0.0057	0.0067	0.0061	0.0062
Final goal distance -- median (m)	0.0051	0.0052	0.0052	0.0052
Final goal distance -- p90 (m)	0.0073	0.0072	0.0071	0.0072
Final goal distance -- worst episode (m)	0.3725	0.5497	0.4180	0.4467
Singularity margin violated (% of steps)	8.05	7.36	7.66	7.69
Joint-limit margin violated (% of steps)	4.15	4.75	3.84	4.25
Collision floor violated (% of steps)	0.00	0.05	0.00	0.02
Episodes that touched the singularity margin (%)	17.20	16.60	17.60	17.13
Episodes that touched the joint-limit margin (%)	5.50	6.30	5.20	5.67
Episodes that touched the collision floor (%)	0.00	0.30	0.10	0.13
Manipulability w -- mean of episode-minimum	0.0729	0.0728	0.0726	0.0728
Manipulability w -- worst single episode	0.0119	0.0230	0.0111	0.0153
Episodic safety cost -- mean	10.39	10.35	9.86	10.20
Episodic safety cost -- p90	3.07	3.11	2.66	2.95

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_ctrl/2026-08-01_05-53-18_ctrl_s54/model_1499.pt

cppo (PPO-Lagrangian, cost_limit = 25)

cppo_s1 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success ≥ 0.50 x commanded goal height (%) <- headline	100.00	99.80	100.00	99.93
... bar reached at any point in the episode (%)	100.00	99.80	100.00	99.93
... legacy fixed-height definition, Day-22 (%)	100.00	99.80	100.00	99.93
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	96.70	98.00	97.60	97.43
Goal-reach success @ 2 cm (%)	99.90	99.70	99.70	99.77
Goal-reach success @ 5 cm (%)	100.00	99.80	100.00	99.93
Final goal distance -- mean (m)	0.0046	0.0053	0.0045	0.0048
Final goal distance -- median (m)	0.0041	0.0043	0.0041	0.0042
Final goal distance -- p90 (m)	0.0063	0.0064	0.0064	0.0064
Final goal distance -- worst episode (m)	0.0322	0.4251	0.0211	0.1595
Singularity margin violated (% of steps)	17.65	19.20	18.41	18.42
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	36.80	39.80	37.30	37.97
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0622	0.0609	0.0616	0.0616
Manipulability w -- worst single episode	0.0000	0.0000	0.0000	0.0000
Episodic safety cost -- mean	19.35	19.78	21.47	20.20
Episodic safety cost -- p90	86.19	93.71	103.80	94.57

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs_l_rl/ur5e_lift_cppo/2026-08-01_01-56-53_cppo_s1/model_1499.pt

cppo_s2 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success $\geq 0.50 \times$ commanded goal height (%) <- headline	99.80	99.80	99.90	99.83
... bar reached at any point in the episode (%)	99.80	99.80	99.90	99.83
... legacy fixed-height definition, Day-22 (%)	99.80	99.80	99.90	99.83
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.60	99.60	99.60	99.60
Goal-reach success @ 2 cm (%)	99.80	99.80	99.90	99.83
Goal-reach success @ 5 cm (%)	99.80	99.80	99.90	99.83
Final goal distance -- mean (m)	0.0045	0.0043	0.0043	0.0044
Final goal distance -- median (m)	0.0035	0.0035	0.0035	0.0035
Final goal distance -- p90 (m)	0.0053	0.0052	0.0052	0.0052
Final goal distance -- worst episode (m)	0.5130	0.4582	0.5514	0.5075
Singularity margin violated (% of steps)	19.83	22.86	21.72	21.47
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	35.10	38.90	37.10	37.03
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.10	0.03
Manipulability w -- mean of episode-minimum	0.0642	0.0634	0.0637	0.0638
Manipulability w -- worst single episode	0.0000	0.0027	0.0004	0.0010
Episodic safety cost -- mean	16.28	16.54	17.82	16.88
Episodic safety cost -- p90	71.06	71.07	74.94	72.36

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_02-08-26_cppo_s2/model_1499.pt

cppo_s3 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	99.90	99.60	99.80
... bar reached at any point in the episode (%)	99.90	99.90	99.60	99.80
... legacy fixed-height definition, Day-22 (%)	99.90	99.90	99.60	99.80
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.40	99.30	98.90	99.20
Goal-reach success @ 2 cm (%)	99.90	99.90	99.60	99.80
Goal-reach success @ 5 cm (%)	99.90	99.90	99.60	99.80
Final goal distance -- mean (m)	0.0040	0.0038	0.0057	0.0045
Final goal distance -- median (m)	0.0033	0.0034	0.0033	0.0033
Final goal distance -- p90 (m)	0.0047	0.0046	0.0048	0.0047
Final goal distance -- worst episode (m)	0.5747	0.3430	0.7537	0.5571
Singularity margin violated (% of steps)	12.74	14.24	12.75	13.24
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	22.00	23.80	19.50	21.77
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0684	0.0679	0.0685	0.0683
Manipulability w -- worst single episode	0.0000	0.0000	0.0000	0.0000
Episodic safety cost -- mean	11.56	12.20	12.37	12.04
Episodic safety cost -- p90	38.48	38.60	45.30	40.79

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_02-19-58_cppo_s3/model_1499.pt

cppo_s4 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success ≥ 0.50 x commanded goal height (%) <- headline	100.00	99.90	100.00	99.97
... bar reached at any point in the episode (%)	100.00	99.90	100.00	99.97
... legacy fixed-height definition, Day-22 (%)	100.00	99.90	100.00	99.97
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	97.10	97.10	96.70	96.97
Goal-reach success @ 2 cm (%)	98.80	98.70	98.60	98.70
Goal-reach success @ 5 cm (%)	100.00	99.80	100.00	99.93
Final goal distance -- mean (m)	0.0039	0.0044	0.0041	0.0041
Final goal distance -- median (m)	0.0033	0.0033	0.0033	0.0033
Final goal distance -- p90 (m)	0.0054	0.0053	0.0057	0.0055
Final goal distance -- worst episode (m)	0.0404	0.4869	0.0496	0.1923
Singularity margin violated (% of steps)	17.14	19.24	17.50	17.96
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	27.00	30.80	27.30	28.37
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0651	0.0640	0.0648	0.0646
Manipulability w -- worst single episode	0.0000	0.0000	0.0000	0.0000
Episodic safety cost -- mean	20.42	21.85	21.90	21.39
Episodic safety cost -- p90	97.30	103.71	114.40	105.14

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_02-31-28_cppo_s4/model_1499.pt

cppo_s5 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success ≥ 0.50 x commanded goal height (%) <- headline	99.80	99.90	99.90	99.87
... bar reached at any point in the episode (%)	99.80	99.90	99.90	99.87
... legacy fixed-height definition, Day-22 (%)	99.80	99.90	99.90	99.87
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	98.00	97.90	98.70	98.20
Goal-reach success @ 2 cm (%)	98.50	99.10	99.30	98.97
Goal-reach success @ 5 cm (%)	99.80	99.90	99.90	99.87
Final goal distance -- mean (m)	0.0051	0.0048	0.0047	0.0049
Final goal distance -- median (m)	0.0040	0.0040	0.0041	0.0040
Final goal distance -- p90 (m)	0.0057	0.0059	0.0056	0.0057
Final goal distance -- worst episode (m)	0.3498	0.3430	0.4168	0.3699
Singularity margin violated (% of steps)	41.93	44.55	44.10	43.53
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	52.70	54.10	56.50	54.43
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0577	0.0570	0.0569	0.0572
Manipulability w -- worst single episode	0.0085	0.0127	0.0090	0.0101
Episodic safety cost -- mean	22.81	24.50	24.24	23.85
Episodic safety cost -- p90	80.62	78.18	82.73	80.51

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_02-43-02_cppo_s5/model_1499.pt

cppo_s50 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	100.00	100.00	99.80	99.93
... bar reached at any point in the episode (%)	100.00	100.00	99.80	99.93
... legacy fixed-height definition, Day-22 (%)	100.00	100.00	99.80	99.93
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	94.30	93.60	93.40	93.77
Goal-reach success @ 2 cm (%)	99.40	99.20	99.00	99.20
Goal-reach success @ 5 cm (%)	100.00	100.00	99.70	99.90
Final goal distance -- mean (m)	0.0057	0.0058	0.0066	0.0060
Final goal distance -- median (m)	0.0053	0.0054	0.0053	0.0053
Final goal distance -- p90 (m)	0.0082	0.0084	0.0083	0.0083
Final goal distance -- worst episode (m)	0.0380	0.0331	0.5671	0.2127
Singularity margin violated (% of steps)	32.91	35.14	31.84	33.30
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	57.00	56.10	53.50	55.53
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0524	0.0511	0.0522	0.0519
Manipulability w -- worst single episode	0.0000	0.0000	0.0000	0.0000
Episodic safety cost -- mean	22.47	25.55	23.63	23.88
Episodic safety cost -- p90	87.71	94.36	89.21	90.43

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_06-04-55_cppo_s50/model_1499.pt

cppo_s51 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	249.8	250.0	250.0	249.9
Lift success ≥ 0.50 x commanded goal height (%) <- headline	99.80	100.00	99.90	99.90
... bar reached at any point in the episode (%)	99.80	100.00	100.00	99.93
... legacy fixed-height definition, Day-22 (%)	99.80	100.00	99.90	99.90
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.40	99.30	99.40	99.37
Goal-reach success @ 2 cm (%)	99.80	100.00	99.90	99.90
Goal-reach success @ 5 cm (%)	99.80	100.00	99.90	99.90
Final goal distance -- mean (m)	0.0055	0.0048	0.0051	0.0051
Final goal distance -- median (m)	0.0047	0.0047	0.0047	0.0047
Final goal distance -- p90 (m)	0.0065	0.0068	0.0064	0.0066
Final goal distance -- worst episode (m)	0.4779	0.0183	0.4180	0.3047
Singularity margin violated (% of steps)	24.39	26.34	27.01	25.91
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	49.40	49.80	49.50	49.57
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0584	0.0581	0.0578	0.0581
Manipulability w -- worst single episode	0.0053	0.0101	0.0071	0.0075
Episodic safety cost -- mean	16.61	17.17	18.55	17.44
Episodic safety cost -- p90	73.84	72.52	74.13	73.50

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_06-16-36_cppo_s51/model_1499.pt

cppo_s52 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	249.8	249.8	249.9
Lift success >= 0.50 x commanded goal height (%) <- headline	100.00	99.90	99.70	99.87
... bar reached at any point in the episode (%)	100.00	99.90	99.70	99.87
... legacy fixed-height definition, Day-22 (%)	100.00	99.90	99.70	99.87
Mean commanded goal height (m)	0.296	0.299	0.301	0.299
Goal-reach success @ 1 cm (%) <- headline	98.80	99.40	98.90	99.03
Goal-reach success @ 2 cm (%)	100.00	99.90	99.70	99.87
Goal-reach success @ 5 cm (%)	100.00	99.90	99.70	99.87
Final goal distance -- mean (m)	0.0044	0.0047	0.0060	0.0050
Final goal distance -- median (m)	0.0043	0.0042	0.0042	0.0042
Final goal distance -- p90 (m)	0.0057	0.0055	0.0056	0.0056
Final goal distance -- worst episode (m)	0.0144	0.4826	0.8073	0.4348
Singularity margin violated (% of steps)	21.69	22.24	23.79	22.57
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	33.20	32.30	34.90	33.47
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0664	0.0673	0.0664	0.0667
Manipulability w -- worst single episode	0.0141	0.0290	0.0128	0.0186
Episodic safety cost -- mean	8.85	8.35	9.54	8.91
Episodic safety cost -- p90	37.16	35.72	39.49	37.46

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_06-28-07_cppo_s52/model_1499.pt

cppo_s53 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	249.8	250.0	250.0	249.9
Lift success $\geq 0.50 \times$ commanded goal height (%) <- headline	99.80	99.90	99.90	99.87
... bar reached at any point in the episode (%)	99.80	99.90	99.90	99.87
... legacy fixed-height definition, Day-22 (%)	99.80	99.90	99.90	99.87
Mean commanded goal height (m)	0.297	0.299	0.300	0.299
Goal-reach success @ 1 cm (%) <- headline	81.60	84.90	80.60	82.37
Goal-reach success @ 2 cm (%)	95.80	96.40	95.50	95.90
Goal-reach success @ 5 cm (%)	99.50	99.80	99.70	99.67
Final goal distance -- mean (m)	0.0078	0.0070	0.0074	0.0074
Final goal distance -- median (m)	0.0047	0.0047	0.0048	0.0047
Final goal distance -- p90 (m)	0.0138	0.0132	0.0140	0.0137
Final goal distance -- worst episode (m)	0.6612	0.4265	0.3775	0.4884
Singularity margin violated (% of steps)	54.77	56.87	54.79	55.48
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	74.30	76.80	73.80	74.97
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0521	0.0509	0.0519	0.0516
Manipulability w -- worst single episode	0.0087	0.0096	0.0060	0.0081
Episodic safety cost -- mean	25.42	26.84	25.70	25.99
Episodic safety cost -- p90	85.70	82.99	83.07	83.92

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_06-39-30_cppo_s53/model_1499.pt

cppo_s54 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	249.8	249.8	250.0	249.9
Lift success >= 0.50 x commanded goal height (%) <- headline	99.70	99.60	100.00	99.77
... bar reached at any point in the episode (%)	99.70	99.70	100.00	99.80
... legacy fixed-height definition, Day-22 (%)	99.70	99.60	100.00	99.77
Mean commanded goal height (m)	0.299	0.295	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.20	98.70	99.10	99.00
Goal-reach success @ 2 cm (%)	99.70	99.60	100.00	99.77
Goal-reach success @ 5 cm (%)	99.70	99.60	100.00	99.77
Final goal distance -- mean (m)	0.0079	0.0083	0.0057	0.0073
Final goal distance -- median (m)	0.0055	0.0055	0.0056	0.0055
Final goal distance -- p90 (m)	0.0073	0.0072	0.0074	0.0073
Final goal distance -- worst episode (m)	0.8966	1.3258	0.0130	0.7451
Singularity margin violated (% of steps)	14.05	12.67	14.04	13.59
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.01	0.01	0.00	0.01
Episodes that touched the singularity margin (%)	19.40	18.80	19.60	19.27
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.30	0.10	0.00	0.13
Manipulability w -- mean of episode-minimum	0.0730	0.0733	0.0730	0.0731
Manipulability w -- worst single episode	0.0000	0.0000	0.0000	0.0000
Episodic safety cost -- mean	13.65	13.01	13.83	13.50
Episodic safety cost -- p90	55.34	40.74	57.11	51.06

checkpoint:

/home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo/2026-08-01_06-51-10_cppo_s54/model_1499.pt

cppo15 (PPO-Lagrangian, cost_limit = 15)

cppo15_s1 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success ≥ 0.50 x commanded goal height (%) <- headline	99.80	100.00	99.60	99.80
... bar reached at any point in the episode (%)	99.80	100.00	99.70	99.83
... legacy fixed-height definition, Day-22 (%)	99.80	100.00	99.60	99.80
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	96.90	98.40	96.50	97.27
Goal-reach success @ 2 cm (%)	99.20	99.90	99.40	99.50
Goal-reach success @ 5 cm (%)	99.80	100.00	99.60	99.80
Final goal distance -- mean (m)	0.0060	0.0050	0.0070	0.0060
Final goal distance -- median (m)	0.0049	0.0049	0.0049	0.0049
Final goal distance -- p90 (m)	0.0070	0.0069	0.0071	0.0070
Final goal distance -- worst episode (m)	0.4723	0.0204	0.5006	0.3311
Singularity margin violated (% of steps)	14.93	16.59	17.05	16.19
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	35.70	35.70	36.30	35.90
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.20	0.07
Manipulability w -- mean of episode-minimum	0.0623	0.0618	0.0619	0.0620
Manipulability w -- worst single episode	0.0000	0.0000	0.0000	0.0000
Episodic safety cost -- mean	15.44	15.35	17.27	16.02
Episodic safety cost -- p90	59.30	69.24	74.04	67.53

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rl_rl/ur5e_lift_cppo15/2026-08-01_17-41-46_cppo15_s1/model_1499.pt

cppo15_s2 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	100.00	99.80	99.80	99.87
... bar reached at any point in the episode (%)	100.00	99.90	99.80	99.90
... legacy fixed-height definition, Day-22 (%)	100.00	99.80	99.80	99.87
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.70	99.40	99.30	99.47
Goal-reach success @ 2 cm (%)	100.00	99.80	99.80	99.87
Goal-reach success @ 5 cm (%)	100.00	99.80	99.80	99.87
Final goal distance -- mean (m)	0.0046	0.0056	0.0052	0.0051
Final goal distance -- median (m)	0.0046	0.0046	0.0046	0.0046
Final goal distance -- p90 (m)	0.0063	0.0062	0.0061	0.0062
Final goal distance -- worst episode (m)	0.0144	0.5497	0.3624	0.3088
Singularity margin violated (% of steps)	15.00	16.01	15.90	15.64
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.13	0.00	0.04
Episodes that touched the singularity margin (%)	38.70	36.80	35.20	36.90
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.20	0.00	0.07
Manipulability w -- mean of episode-minimum	0.0640	0.0643	0.0645	0.0643
Manipulability w -- worst single episode	0.0117	0.0226	0.0124	0.0156
Episodic safety cost -- mean	4.83	4.59	5.16	4.86
Episodic safety cost -- p90	14.00	12.89	15.02	13.97

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs1_rl/ur5e_lift_cppo15/2026-08-01_17-53-14_cppo15_s2
/model_1499.pt

cppo15_s3 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	99.90	99.80	99.87
... bar reached at any point in the episode (%)	100.00	99.90	99.80	99.90
... legacy fixed-height definition, Day-22 (%)	99.90	99.90	99.80	99.87
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.70	99.40	99.40	99.50
Goal-reach success @ 2 cm (%)	99.90	99.90	99.80	99.87
Goal-reach success @ 5 cm (%)	99.90	99.90	99.80	99.87
Final goal distance -- mean (m)	0.0047	0.0050	0.0051	0.0049
Final goal distance -- median (m)	0.0042	0.0042	0.0041	0.0042
Final goal distance -- p90 (m)	0.0051	0.0051	0.0051	0.0051
Final goal distance -- worst episode (m)	0.4619	0.7081	0.5646	0.5782
Singularity margin violated (% of steps)	24.42	25.44	24.92	24.93
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	38.80	39.50	38.10	38.80
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.10	0.03
Manipulability w -- mean of episode-minimum	0.0629	0.0622	0.0636	0.0629
Manipulability w -- worst single episode	0.0000	0.0225	0.0174	0.0133
Episodic safety cost -- mean	10.67	11.81	10.98	11.15
Episodic safety cost -- p90	43.32	50.06	47.12	46.83

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo15/2026-08-01_18-05-05_cppo15_s3
/model_1499.pt

cppo15_s4 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success ≥ 0.50 x commanded goal height (%) <- headline	100.00	100.00	99.90	99.97
... bar reached at any point in the episode (%)	100.00	100.00	99.90	99.97
... legacy fixed-height definition, Day-22 (%)	100.00	100.00	99.90	99.97
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	98.60	99.50	99.00	99.03
Goal-reach success @ 2 cm (%)	100.00	100.00	99.90	99.97
Goal-reach success @ 5 cm (%)	100.00	100.00	99.90	99.97
Final goal distance -- mean (m)	0.0039	0.0038	0.0043	0.0040
Final goal distance -- median (m)	0.0037	0.0038	0.0038	0.0038
Final goal distance -- p90 (m)	0.0053	0.0052	0.0052	0.0052
Final goal distance -- worst episode (m)	0.0186	0.0115	0.3971	0.1424
Singularity margin violated (% of steps)	11.89	13.04	13.12	12.68
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	29.70	28.10	30.30	29.37
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0660	0.0658	0.0660	0.0659
Manipulability w -- worst single episode	0.0247	0.0302	0.0315	0.0288
Episodic safety cost -- mean	3.05	3.60	3.65	3.43
Episodic safety cost -- p90	8.67	11.52	11.74	10.64

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs1_rl/ur5e_lift_cppo15/2026-08-01_18-18-01_cppo15_s4
/model_1499.pt

cppo15_s5 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	100.00	100.00	99.97
... bar reached at any point in the episode (%)	99.90	100.00	100.00	99.97
... legacy fixed-height definition, Day-22 (%)	99.90	100.00	100.00	99.97
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.70	99.80	99.70	99.73
Goal-reach success @ 2 cm (%)	99.90	100.00	100.00	99.97
Goal-reach success @ 5 cm (%)	99.90	100.00	100.00	99.97
Final goal distance -- mean (m)	0.0051	0.0045	0.0044	0.0047
Final goal distance -- median (m)	0.0042	0.0043	0.0043	0.0043
Final goal distance -- p90 (m)	0.0057	0.0058	0.0057	0.0057
Final goal distance -- worst episode (m)	0.6484	0.0124	0.0117	0.2242
Singularity margin violated (% of steps)	40.17	41.12	41.92	41.07
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	54.80	55.20	56.10	55.37
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0581	0.0576	0.0577	0.0578
Manipulability w -- worst single episode	0.0209	0.0266	0.0211	0.0229
Episodic safety cost -- mean	14.17	14.13	15.16	14.49
Episodic safety cost -- p90	46.82	45.86	51.17	47.95

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rsl_rl/ur5e_lift_cppo15/2026-08-01_18-30-16_cppo15_s5/model_1499.pt

cppo15_s50 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.80	100.00	99.70	99.83
... bar reached at any point in the episode (%)	99.80	100.00	99.70	99.83
... legacy fixed-height definition, Day-22 (%)	99.80	100.00	99.70	99.83
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.60	99.70	99.10	99.47
Goal-reach success @ 2 cm (%)	99.80	100.00	99.70	99.83
Goal-reach success @ 5 cm (%)	99.80	100.00	99.70	99.83
Final goal distance -- mean (m)	0.0050	0.0041	0.0052	0.0048
Final goal distance -- median (m)	0.0040	0.0041	0.0040	0.0040
Final goal distance -- p90 (m)	0.0057	0.0058	0.0059	0.0058
Final goal distance -- worst episode (m)	0.5850	0.0122	0.5832	0.3935
Singularity margin violated (% of steps)	19.77	22.61	21.28	21.22
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	40.30	42.90	40.70	41.30
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.10	0.03
Manipulability w -- mean of episode-minimum	0.0635	0.0626	0.0635	0.0632
Manipulability w -- worst single episode	0.0131	0.0162	0.0125	0.0139
Episodic safety cost -- mean	10.79	10.60	11.66	11.02
Episodic safety cost -- p90	44.97	39.00	49.22	44.40

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs1_rl/ur5e_lift_cppo15/2026-08-01_18-41-20_cppo15_s50/model_1499.pt

cppo15_s51 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	99.80	99.60	99.77
... bar reached at any point in the episode (%)	99.90	99.90	99.70	99.83
... legacy fixed-height definition, Day-22 (%)	99.90	99.80	99.60	99.77
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	98.30	98.20	98.70	98.40
Goal-reach success @ 2 cm (%)	99.90	99.80	99.60	99.77
Goal-reach success @ 5 cm (%)	99.90	99.80	99.60	99.77
Final goal distance -- mean (m)	0.0051	0.0055	0.0064	0.0057
Final goal distance -- median (m)	0.0046	0.0047	0.0047	0.0047
Final goal distance -- p90 (m)	0.0063	0.0063	0.0062	0.0063
Final goal distance -- worst episode (m)	0.3862	0.4690	0.6163	0.4905
Singularity margin violated (% of steps)	20.20	23.36	21.55	21.70
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	26.20	29.00	27.40	27.53
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0685	0.0678	0.0679	0.0681
Manipulability w -- worst single episode	0.0102	0.0130	0.0103	0.0112
Episodic safety cost -- mean	14.78	15.26	15.85	15.30
Episodic safety cost -- p90	67.80	65.51	69.91	67.74

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs1_rl/ur5e_lift_cppo15/2026-08-01_18-52-32_cppo15_s51/model_1499.pt

cppo15_s52 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	100.00	100.00	100.00	100.00
... bar reached at any point in the episode (%)	100.00	100.00	100.00	100.00
... legacy fixed-height definition, Day-22 (%)	100.00	100.00	100.00	100.00
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.40	99.50	99.40	99.43
Goal-reach success @ 2 cm (%)	100.00	100.00	100.00	100.00
Goal-reach success @ 5 cm (%)	100.00	100.00	100.00	100.00
Final goal distance -- mean (m)	0.0041	0.0041	0.0041	0.0041
Final goal distance -- median (m)	0.0039	0.0040	0.0040	0.0040
Final goal distance -- p90 (m)	0.0055	0.0056	0.0054	0.0055
Final goal distance -- worst episode (m)	0.0119	0.0126	0.0130	0.0125
Singularity margin violated (% of steps)	29.74	28.25	27.78	28.59
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	46.70	44.80	42.70	44.73
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0613	0.0615	0.0620	0.0616
Manipulability w -- worst single episode	0.0227	0.0235	0.0267	0.0243
Episodic safety cost -- mean	10.24	11.25	10.39	10.63
Episodic safety cost -- p90	42.96	44.78	42.84	43.53

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs1_rl/ur5e_lift_cppo15/2026-08-01_19-04-15_cppo15_s52/model_1499.pt

cppo15_s53 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	99.90	99.90	100.00	99.93
... bar reached at any point in the episode (%)	99.90	99.90	100.00	99.93
... legacy fixed-height definition, Day-22 (%)	99.90	99.90	100.00	99.93
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.60	99.30	99.60	99.50
Goal-reach success @ 2 cm (%)	99.90	99.90	100.00	99.93
Goal-reach success @ 5 cm (%)	99.90	99.90	100.00	99.93
Final goal distance -- mean (m)	0.0044	0.0047	0.0040	0.0044
Final goal distance -- median (m)	0.0039	0.0040	0.0038	0.0039
Final goal distance -- p90 (m)	0.0059	0.0058	0.0057	0.0058
Final goal distance -- worst episode (m)	0.3387	0.6211	0.0151	0.3250
Singularity margin violated (% of steps)	11.02	11.96	11.56	11.51
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	26.60	27.20	26.70	26.83
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0671	0.0669	0.0674	0.0671
Manipulability w -- worst single episode	0.0106	0.0182	0.0108	0.0132
Episodic safety cost -- mean	7.59	7.27	8.25	7.70
Episodic safety cost -- p90	21.41	18.48	18.94	19.61

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs1_rl/ur5e_lift_cppo15/2026-08-01_19-15-27_cppo15_s53/model_1499.pt

cppo15_s54 -- 1000 episodes per eval seed

Metric	eval-seed 101	eval-seed 102	eval-seed 103	mean of 3
Mean episode length (steps)	250.0	250.0	250.0	250.0
Lift success >= 0.50 x commanded goal height (%) <- headline	100.00	99.80	99.80	99.87
... bar reached at any point in the episode (%)	100.00	99.80	99.80	99.87
... legacy fixed-height definition, Day-22 (%)	100.00	99.80	99.80	99.87
Mean commanded goal height (m)	0.296	0.299	0.300	0.298
Goal-reach success @ 1 cm (%) <- headline	99.60	99.40	99.20	99.40
Goal-reach success @ 2 cm (%)	100.00	99.80	99.80	99.87
Goal-reach success @ 5 cm (%)	100.00	99.80	99.80	99.87
Final goal distance -- mean (m)	0.0042	0.0049	0.0050	0.0047
Final goal distance -- median (m)	0.0041	0.0040	0.0040	0.0040
Final goal distance -- p90 (m)	0.0059	0.0058	0.0060	0.0059
Final goal distance -- worst episode (m)	0.0119	0.4177	0.5671	0.3322
Singularity margin violated (% of steps)	34.73	33.31	36.97	35.00
Joint-limit margin violated (% of steps)	0.00	0.00	0.00	0.00
Collision floor violated (% of steps)	0.00	0.00	0.00	0.00
Episodes that touched the singularity margin (%)	41.20	38.90	42.80	40.97
Episodes that touched the joint-limit margin (%)	0.00	0.00	0.00	0.00
Episodes that touched the collision floor (%)	0.00	0.00	0.00	0.00
Manipulability w -- mean of episode-minimum	0.0615	0.0625	0.0612	0.0617
Manipulability w -- worst single episode	0.0143	0.0203	0.0165	0.0170
Episodic safety cost -- mean	15.51	13.46	16.91	15.29
Episodic safety cost -- p90	57.88	50.38	60.67	56.31

checkpoint: /home/mte/Abdur_Rabbi_THESIS/Comparison_test/logs/rs1_rl/ur5e_lift_cppo15/2026-08-01_19-27-01_cppo15_s54/model_1499.pt